

SIOUX FALLS CLIMATE, SD, USA

WMO: 711680

Lat: 43.7336N

Lon: 96.6233W

Elev: 1580

StdP: 13.88

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -11.0	(c) -6.8	(d) -16.3	(e) 2.4	(f) -10.7	(g) -12.3	(h) 3.0	(i) -6.8	(j) 26.4	(k) 23.1	(l) 24.4	(m) 21.4	(n) 7.4	(o) 280	(p) 0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 17.2	(c) 88.4	(d) 73.4	(e) 85.6	(f) 71.8	(g) 83.0	(h) 70.4	(i) 76.7	(j) 84.8	(k) 74.6	(l) 82.8	(m) 72.7	(n) 80.5	(o) 10.7	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.2	135.6	82.5	72.0	125.4	80.6	70.0	116.9	78.3	41.4	85.1	39.3	82.4	37.5	80.8	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 24.5	(b) 20.9	(c) 18.8	(e) -19.1	(f) 91.9	(g) 7.5	(h) 3.6	(i) -24.5	(j) 94.5	(k) -28.9	(l) 96.6	(m) -33.1	(n) 98.7	(o) -38.6	(p) 101.3		
(a) 24.5	(b) 20.9	(c) 18.8	(e) -20.4	(f) 79.6	(g) 7.5	(h) 1.7	(i) -25.8	(j) 80.9	(k) -30.2	(l) 81.9	(m) -34.4	(n) 82.8	(o) -39.8	(p) 84.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.9	15.5	17.5	32.2	44.6	57.5	68.6	71.8	68.5	61.2	46.4	33.7	19.1	
	DBStd	22.39	13.09	12.42	13.10	10.29	9.13	6.45	5.78	5.60	7.81	9.16	11.17	11.57	
	HDD50	4440	1069	910	565	218	35	0	0	0	9	179	497	958	
	HDD65	7940	1534	1330	1016	615	265	36	12	29	163	577	940	1423	
	CDD50	2567	0	0	14	55	268	559	674	575	345	69	8	0	
	CDD65	589	0	0	1	2	33	144	221	138	48	2	0	0	
	CDH74	4863	0	0	3	43	366	1177	1857	1029	367	21	0	0	
	CDH80	1319	0	0	0	9	97	348	561	236	66	2	0	0	
Wind	WSAvg	8.9	9.0	9.4	10.4	11.3	9.3	8.6	7.1	6.7	8.0	9.2	9.4	9.0	
Precipitation	PrecAvg	27.4	0.6	0.8	1.5	2.9	3.7	4.4	3.3	3.3	2.7	2.2	1.2	0.8	
	PrecMax	37.1	1.4	2.2	4.1	6.8	8.0	12.9	8.3	7.8	7.3	6.1	4.3	2.1	
	PrecMin	17.5	0.0	0.2	0.2	0.5	1.0	1.0	0.4	1.1	0.2	0.3	0.0	0.1	
	PrecStd	5.0	0.3	0.5	1.0	1.4	1.9	2.2	2.2	1.9	1.8	1.7	1.2	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.3	55.2	71.2	80.6	87.7	91.9	92.7	89.1	85.7	76.8	64.9	48.6	
		MCWB	37.2	47.8	58.6	59.8	63.4	72.4	75.6	75.0	71.5	61.0	54.5	44.0	
	2%	DB	38.6	42.9	63.1	71.5	82.2	87.4	88.8	85.3	82.0	70.8	59.2	41.7	
		MCWB	34.9	38.5	51.9	55.8	64.3	70.5	76.2	74.2	69.1	57.4	50.4	36.5	
	5%	DB	35.4	38.0	57.3	66.1	78.0	83.9	85.9	82.6	78.2	65.9	53.6	37.4	
		MCWB	33.3	35.1	49.9	52.6	62.4	68.6	74.3	72.3	67.2	55.0	45.9	35.0	
	10%	DB	33.1	34.6	51.2	61.1	73.2	80.8	82.9	79.8	75.0	61.3	49.4	34.3	
		MCWB	31.5	32.3	45.2	49.6	59.8	67.4	72.1	70.4	65.3	52.3	44.5	32.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.3	48.8	60.6	60.9	70.1	76.9	80.2	78.2	74.2	63.1	55.7	44.4
			MCDB	42.6	55.5	67.3	74.6	78.9	87.9	88.6	85.9	82.0	72.5	62.6	47.5
2%		WB	35.3	39.3	54.1	58.4	66.9	73.8	77.5	75.7	71.4	59.2	50.9	37.4	
		MCDB	38.0	42.3	60.2	69.5	77.8	83.6	85.0	82.8	79.1	67.8	57.8	40.1	
5%		WB	33.5	35.3	49.7	54.9	64.5	71.7	75.5	73.5	69.5	56.5	47.4	35.0	
		MCDB	35.3	38.0	57.1	63.4	75.2	81.0	84.2	80.1	75.6	64.6	52.8	37.3	
10%		WB	31.7	32.5	45.2	51.5	61.9	69.2	73.4	71.5	67.1	53.1	44.1	32.6	
		MCDB	33.3	34.3	51.9	58.8	70.2	77.7	81.6	78.3	72.7	59.9	49.4	34.4	
Mean Daily Temperature Range	5% DB	MDBR	14.4	14.6	15.7	18.1	18.3	16.8	17.2	17.3	18.7	17.1	15.9	13.8	
		MCDBR	16.4	18.3	24.5	27.2	24.7	21.2	18.8	19.8	21.8	24.6	22.0	16.0	
		MCWBR	14.0	13.7	14.9	14.2	11.0	9.8	8.9	10.5	10.7	13.4	13.4	12.5	
	5% WB	MCDBR	15.2	17.5	22.6	23.3	20.6	18.6	17.6	17.7	17.8	20.6	20.0	15.2	
		MCWBR	13.0	13.3	14.3	13.9	10.8	10.7	9.4	10.6	10.9	12.6	13.1	12.2	
Clear-Sky Solar Irradiance	taub	0.269	0.282	0.317	0.368	0.384	0.394	0.401	0.403	0.355	0.325	0.292	0.272		
	taud	2.431	2.416	2.409	2.291	2.323	2.351	2.332	2.299	2.412	2.472	2.482	2.439		
	Ebn at Noon	277	292	293	284	281	278	275	270	276	272	266	264		
	Edh at Noon	23	28	32	40	40	39	39	39	32	26	22	21		
All-Sky Solar Radiation	RadAvg	601	936	1222	1540	1774	1987	2113	1745	1388	937	645	478		
	RadStd	55	74	100	141	130	129	78	134	103	112	46	40		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page